## 3.9 Shear Force And Bending Moment Diagrams For A Simply Supported Beam

### 3.9.1 A Simply Supported Beam With A Point Load At Mid-Point

Fig. 3.10 (a) shows a beam AB of length L simply supported at the ends A and B and carrying a point load W at its middle point C. Since the beam is loaded symmetrically the reactions at the support will be equal to $\frac{W}{2}$.

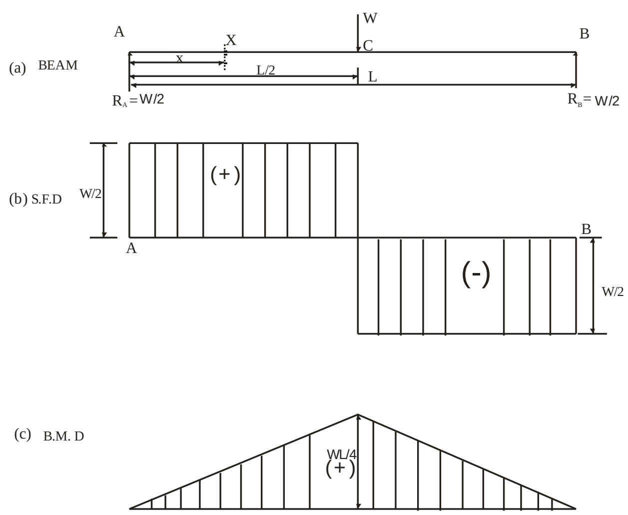


Fig. 3.10

Hence $R_A = R_B = \frac{\quad}{2}$.

**S.F.D.:** For the section AC let us take a section X at a distance x from the end A between A and C. Here we have considered the left portion of the section. The shear force at X will be equal to the resultant force acting on the left portion of the section. But the resultant force on the left portion is $\frac{W}{2}$ acting upwards. According to the sign convention, the resultant force on the left portion acting upwards is considered positive. Hence shear force at X is positive and its magnitude is $\frac{W}{2}$.

$$\therefore F_x = +\frac{W}{2} \quad \text{(Constant)}$$

For the section CB let us take a section X at a distance x from the end A. The resultant force on the left portion will be

$$\left(\frac{W}{2} - W\right) = -\frac{W}{2}$$

This force will also remain constant between C and B. Hence shear force between C and B is equal to $-\frac{W}{2}$.

At the section C the shear force changes from $+\frac{W}{2}$ to $-\frac{W}{2}$.

The shear force diagram is shown in Fig. 3.10 (b).

**B.M. D:**

Section AC, $M_x = R_A \,.\, x$

$$\text{or} \qquad M_x = +\frac{W}{2} \,.\, x \text{ (Linear)} \qquad \text{.....(i)}$$

B.M. will be positive as for the left portion of the section, the moment of all forces at X is clockwise. Moreover, the bending of beam takes place in such a manner that concavity is at the top of the beam.

At A, $x = 0$ Hence $M_A = \frac{W}{2} \times 0 = 0$

At C, $x = \frac{L}{2}$ hence $M_C = \frac{W}{2} \times \frac{L}{2} = \frac{W \times L}{4}$

Section CB, $M_x = R_A \,.\, x - W \times \left(x - \frac{L}{2}\right)$

$$= \frac{W}{2} \,.\, x - Wx + W \times \frac{L}{2} = \frac{WL}{2} - \frac{Wx}{2} \text{ (Linear)}$$

At C, $x = \frac{L}{2}$ $\therefore M_C = \frac{WL}{2} - \frac{W}{2} \times \frac{L}{2} = \frac{W \times L}{4}$

At B, $x = L$ $\therefore M_B = \frac{WL}{2} - \frac{W}{2} \times L = 0$

The B.M. diagram can be completed as shown in Fig. 3.10 (c).

It is to be worth noted the bending moment is maximum at the middle point C, where the shear force changes its sign.

### 3.9.2 A Simply Supported Beam With An Eccentric Point Load

Let us first calculate the reactions, by taking moments about A or about B.

Taking moments of the forces on the beam about A, we get

$$R_B \times L = W \times a$$

$$\therefore \quad R_B = \frac{W.a}{L}$$

and $$R_A = W - R_B = W - \frac{W.a}{L}$$

$$= W\left(1 - \frac{a}{L}\right) = W\left(\frac{L-a}{L}\right) = \frac{W \times b}{L} \qquad (\because L = a + b)$$

**S.F.D.:** For the section AC let us consider a section *X* at distance *x* from the end A. The shear force $F_x$ at the section is given by,

$$F_x = +R_A = +\frac{W.b}{L} \quad \text{(Constant)}$$

The shear force will be positive as the resultant force on the left portion of the section is acting upwards.

For the section CB let us consider a section X at a distance *x* from the end A. The resultant force on the left portion will be $R_A - W$.

or $$\frac{W.a}{L} - W = W.\left(\frac{b-L}{L}\right) = -W\left(\frac{L-b}{L}\right) = -\frac{W.a}{L} \quad \text{(Constant)}$$

At the section C, the shear force changes from $\frac{W.b}{L}$ to $-\frac{W.a}{L}$. The shear force diagram is shown in Fig 3.11 (b).

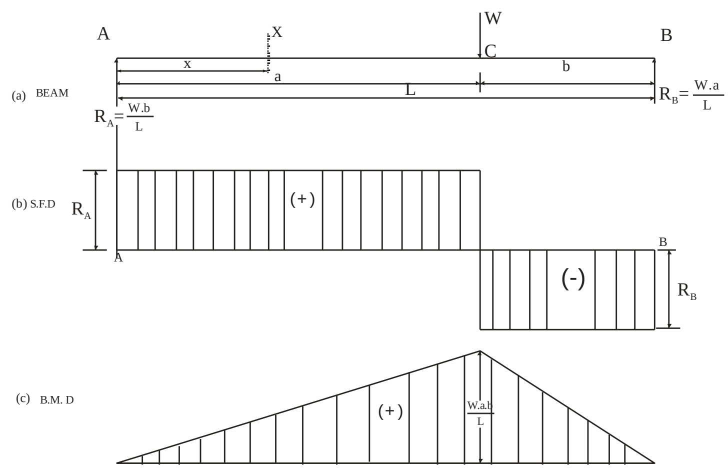


Fig. 3.11

**B.M.D.**

Section AC, $M_x = R_A \cdot x$

$= + \frac{W.b}{L} \cdot x$ (Linear) (Plus sign due to sagging)

At A, $x = 0$ Hence $M_A = \frac{W.b}{L} \times 0 = 0$

At C, $x = a$ Hence $M_C = \frac{W.b}{L} .a = \frac{W.a.b}{L}$

The B.M. is zero at B. Hence B.M. will decrease from $\frac{W.a.b}{L}$ at C to zero at B following a straight line law. The B.M. diagram is drawn in Fig. 3.11 (c).

It is to be worth noted from the shear force and bending moment diagrams, the B.M. is maximum at C where the S.F. changes its sign.

### 3.9.3 A Simply Supported Beam Carrying A Uniformly Distributed Load

Fig. 3.12 shows a beam AB of length L simply supported at the ends A and B and carrying a uniformly distributed load of *w* per unit length over the entire length. The reactions at the supports will be equal and their magnitude will be half the total load on the entire length.

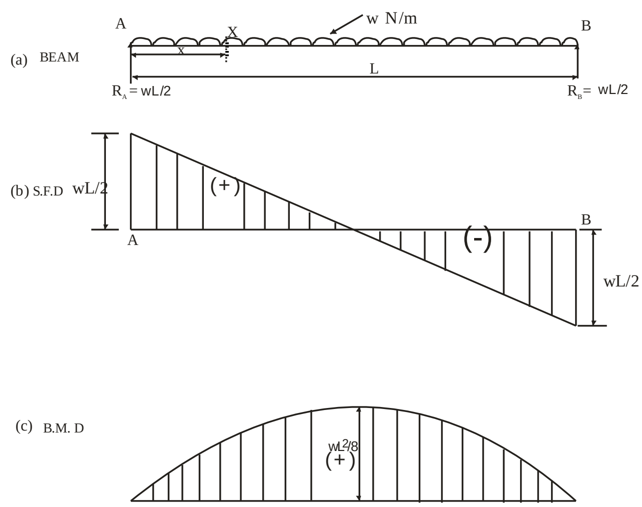


Fig. 3.12

$\therefore \quad R_A = R_B = \frac{}{2}$

**S.F.D.:** For the section AC let us consider a section *X* at distance *x* from the end A. The shear force $F_x$ at the section is given by,

$$F_x = + R_A - w.x = + \frac{w.L}{2} - w.x \text{ (Linear)} \qquad \text{....(i)}$$

The values of shear force at different points are :

At A, $x = 0 \quad \therefore F_A = + \frac{w.L}{2} - \frac{w.0}{2} = + \frac{w.L}{2}$

At B, $x = L \quad \therefore F_B = + \frac{w.L}{2} - w \cdot L = - \frac{w.L}{2}$

At C, $x = \frac{L}{2} \quad \therefore F_C = + \frac{w.L}{2} - w. \frac{L}{2} = 0$

The shear force diagram is drawn as shown in Fig. 3.12 (b).

**B.M.D.**

Section AC, $M_x = + R_A \,.\, x - w \,.\, x \,.\, \frac{x}{2}$

$$= \frac{w.L}{2}.x - \frac{w.x^2}{2} \quad \text{(Parabolic)} \qquad \left(\because \quad R_A = \frac{w.L}{2}\right) \qquad \text{.....(ii)}$$

The values of B.M. at different points are :

At A, $x = 0$ $\quad \therefore\ M_A = \frac{w.L}{2}.0 - \frac{w.0}{2} = 0$

At B, $x = L$ $\quad \therefore\ M_B = \frac{w.L}{2}.L - \frac{w}{2}.L^2 = 0$

At C, $x = \frac{L}{2}$ $\quad \therefore\ M_C = \frac{w.L}{2}.\frac{L}{2} - \frac{w}{2}.\left(\frac{L}{2}\right)^2$

$$= \frac{w.L^2}{4} - \frac{w.L^2}{8} = +\frac{w.L^2}{8}$$

Thus the B.M. increases according to parabolic law from zero at A to $+\frac{w.L^2}{8}$ at the middle point of the beam and from this value the B.M. decreases to zero at B according to the parabolic law.

The B.M. diagram is drawn as shown in Fig. 3.12(c).

**3.9.4 A Simply Supported Beam With Combination of Loads (U.D.L. And Point loads)**

Let us first calculate the reactions, by taking moments about A or about B.

Let us take moments of the forces on the beam about A.

$R_B \times 8 = 6 \times 6 + 2 \times 2 - 2 \times 2 \times 6$

($2 \times 2 \times 6$ is the moment of udl about support A. The udl is converted into point load (Fig. 3.12 (d)) and applied at C.G. of the udl.)

$\therefore \qquad R_B = 8$ kN

Similarly taking moments of the forces on the beam about B.

$R_A \times 8 = 2 \times 6 + 6 \times 2 - 2 \times 2 \times 2$

($2 \times 2 \times 2$ is the moment of udl about support B. The udl is converted into point load (Fig. 3.13 (d)) and applied at C.G. of the udl.)

$\therefore \quad R_A = 4$ kN

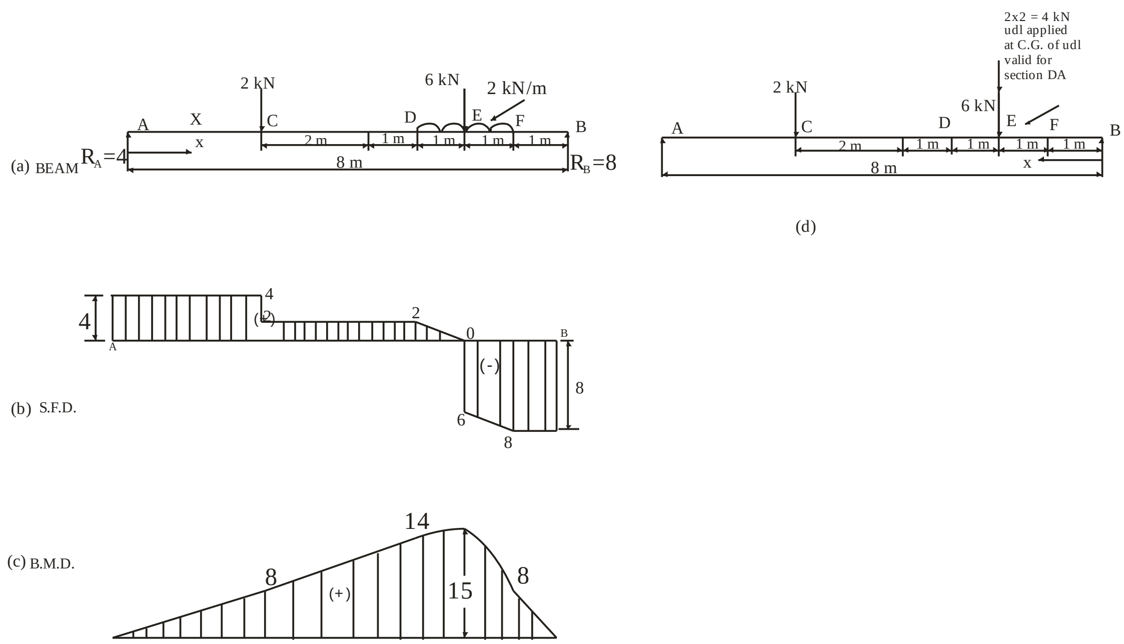


Fig. 3.13

**S.F.D.:** For the section AC let us consider a section *X* at distance *x* from the end A. The shear force $F_x$ at the section is given by,

$$F_x = +R_A = +4 \text{ kN (Constant)}$$

The shear force will be positive as the resultant force on the left portion of the section is acting upwards.

Section CD, $\quad F_x = +4 - 2 = 2$ kN (Constant)

Section DE, $\quad F_x = +4 - 2 - 2(x-5)$ (Linear)

At D, $x = 5 \ \therefore F_D = 2$ kN

At E, $x = 6 \ \therefore F_E = 0$ kN

Section EF, $\quad F_x = +4 - 2 - 2(x-5) - 6$ (Linear)

At E, x = 6 $\therefore F_E$ = -6 kN (Sudden change due to point load)

At F, x = 7 $\therefore F_F$ = 8 kN (constant)

Section FB, $F_x$ = +4 -2 – 2x2 – 6

= 8 kN (Constant)

At F, x = 7 $\therefore F_F$ = -8 kN

At B, x = 8 $\therefore F_B$ = -8 kN

The shear force diagram is shown in Fig 3.13 (b).

**B.M.D.**

Section AC, $M_x = R_A . x$ (Linear)

$= + 4 . x$ (Linear) (Plus sign due to sagging)

At A, $x = 0$ $\therefore M_A = 4 \times 0 = 0$

At C, $x = 2$ $\therefore M_C = 4 \times 2$ = 8 kN-m

Section CD, $M_x = + 4 . x - 2 .(x-2)$ (Linear)

At C, $x = 2$ $\therefore M_C$ = 8 kN-m

At D, $x = 5$ $\therefore M_D$ = 14 kN-m

Section DE, $M_x = + 4 . x - 2 .(x-2) - \frac{2}{2}(x-5)^2$ (Parabolic)

At D, $x = 5$ $\therefore M_D$ = 14 kN-m

At E, $x = 6$ $\therefore M_E$ = 15 kN-m

Section EF, $M_x = + 4 . x - 2 .(x-2) - \frac{2}{2}(x-5)^2 - 6 (x-6)$ (Parabolic)

At E, $x = 6$ $\therefore M_E$ = 15 kN-m

At F, $x = 7$ $\therefore M_F$ = 8 kN-m

For section FB, This section is out of the udl. Hence the udl is to be converted into equivalent point load. The total udl (2x2) is assumed to be applied at C.G. of the udl i.e. 2 m away from the support B as shown in Fig. 3.13 (d). To draw BMD for section FB Fig. 3.13 (d) is to be considered.

Section FB, $M_x = +4 \cdot x - 2 \cdot (x-2) - 4 \cdot (x-6) - 6 \cdot (x-6)$ (Linear)

At F, $x = 7$ $\therefore$ $M_F = 8$ kN-m

At B, $x = 8$ $\therefore$ $M_B = 0$

The B.M. is zero at B. The B.M. diagram is drawn in Fig. 3.13 (c).

It is to be worth noted from the shear force and bending moment diagrams, the B.M. is maximum at E where the S.F. changes its sign.